

Source data under scrutiny



The source data underlying results and figures in our manuscripts can be critical for assessing the rigour of our studies and for reproducibility. Ensuring the quality and provenance of these data during manuscript preparation is essential bookkeeping.

Nature Portfolio journals have been mandating or strongly recommending the sharing of the source data underlying the figures published in our papers for well over a decade. Much more than just underlying a pretty figure, these data represent the results of the study, can give real insight into statistical handling of data and data quality and, in rare cases, can highlight problematic irregularities in experimental procedures and evidence.

Historically, shared source data were often left in the dark: omnipresent, but only meaningfully examined by the most interested of readers. However, the rise of machine learning

models has made source data readily accessible and straightforward to mine for discrepancies between the data and the figures.

A recent spate of high-profile corrections and retractions launched a wave of source data corrections even within our journal, in which authors have taken the initiative to flag and patch inconsistencies between figures and source data in their published papers. In the vast majority of cases, the issues identified arise from unintentional copy-and-paste errors or updates to figures made during revision.

While these issues can readily be addressed to correct the version of record and in most cases do not reflect problems with the overall conclusions of the article, these errors – even when benign – can cast doubt on the validity of a study.

We urge our authors, new and experienced alike, to remember that when papers are accepted-in-principle we will call for source data, as we have done since our first issue nearly 10 years ago. We strongly recommend that authors be especially mindful of bookkeeping during manuscript preparation and

pin their underlying source data to figures throughout versioning. We also recommend that our authors, especially corresponding authors, spend time assessing source data before hitting ‘submit’. Moreover, the same machine learning models that are being used to hunt discrepancies in published source data can be used to pre-emptively. Ultimately, spending a proportionally trivial amount of time during figure preparation and source data quality control can save a lot of hassle and unnecessary speculation about a paper, and more broadly about a research team’s integrity, in the long run.

We value data sharing in its many forms (*Nat. Biomed. Eng.* **4**, 845–846; 2020). From our editorial perspective, we see very plainly that source data are invisible no more. While we are actively discussing ways in which we can be more proactive in assessing source data, we hope that encouraging good bookkeeping habits for our authors can help future-proof our studies and ensure the quality of the papers in our pages.

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